Fabrizio Romanelli

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About

Senior Technical Leader and Research Engineer with 20 years of sustained impact spanning industrial robotics, physics-based simulation, and deep learning. Expert in bridging the sim-to-real gap for complex electromechanical systems by leveraging advanced visual perception, synthetic sensor data generation, and World Models. Proven track record in designing real-time robot control frameworks, multibody kinematics, and contact-rich collision-avoidance algorithms within production engineering environments (Comau, IIT, AlmavivA). Technical vision leader and experienced software manager adept at coordinating cross-functional teams of 10+ people, deploying scalable AI pipelines, and architecting advanced simulation-driven workflows for next-generation intelligent systems.

Education

PhD Computer Science, Control and Geoinformation – University of Rome Tor Vergata	2020 – 2024
MSc Computer Science (with distinction) – University of Rome – Tor Vergata [with distinction, 110/110 cum laude]	2003 – 2005
BSc Computer Science – University of Rome – Tor Vergata	1998 - 2003

Professional experiences

AI & Robotics Engineer – AlmavivA — 05.2024 – present

Architected and designed project proposals integrating World Models and generative AI to systematically predict physical behaviors and close the sim-to-real gap in autonomous systems.
Designed and developed industry-grade AI solutions for advanced visual object detection (YOLOv8) and robust perception in highly cluttered, non-deterministic operational environments.
Engineered and deployed scalable, enterprise-level solutions leveraging Large Language Models (LLMs) to extract automated project insights, requirements, and technical risk analysis.
Served as a technical mentor for engineering teams, driving best practices in object-oriented programming (OOP), software versioning, and agile deployment pipelines.

Senior Software Engineer – Agrivol — 06.2021 – 05.2024

Developed production-grade software stacks for agricultural robotics, utilizing deep learning techniques for high-fidelity synthetic sensor data generation and real-time image segmentation.
Engineered resilient navigation algorithms capable of handling highly variable physical terrains and dynamic environmental contacts.
Collaborated directly on corporate business strategy, mapping technical simulation and AI capabilities to long-term commercial product roadmaps.

Researcher – CNR-ISTP — 12.2022 – 05.2024

Awarded a research fellowship focused on the Electron Cyclotron Resonance Heating (ECRH) system for the DTT (Divertor Tokamak Test facility) international nuclear fusion project.
Designed the complex control architecture, real-time remote monitoring systems, and automated power supply management systems for high-power gyrotrons.

Training Specialist on industrial robotics – IN.SI. — 03.2021 – 12.2023

Conducted advanced technical training on industrial robot control systems, rigid-body kinematics, multi-body dynamics, and fieldbus-based PLC interfacing.
Developed high-fidelity interactive simulations to model, test, and program complex kinematics workflows for Universal Robots and ABB platforms.

Senior Cloud Engineer – Applaudart 03.2019 – 01.2021

Led the core backend development team architecting high-performance cloud computing systems on Amazon Web Services (AWS), utilizing Lambda, DynamoDB, ElasticSearch, and AppSync.

Specialized in the integration of highly complex, enterprise-class distributed systems with automated CI/CD pipelines to ensure scalable out-of-simulation workflows.

Robotics Specialist – University of Rome Tor Vergata 01.2020 – 05.2020

Designed a distributed, low-latency ROS 2 software architecture optimized for resilient perception in autonomous mobile systems.

Investigated state-of-the-art VSLAM and deep learning methodologies for physics-based synthetic sensor data generation.

Developed high-level C++ wrappers for RGB-D camera drivers, integrating and benchmarking ORBSLAM2 optimization models across monocular, stereo, and depth-sensing arrays in real-world environments.

Senior Software Engineer – Istituto Italiano di Tecnologia 12.2017 – 02.2019

Core member of the Advanced Robotics Research Line; supervised engineering teams and postgraduate students in developing real-time control frameworks for legged robots.

Designed modular software architectures for robotic locomotion and multi-DoF manipulation, ensuring high-predictive accuracy during contact-rich physical interactions.

Managed and maintained advanced CI/CD regression testing pipelines on GitLab; integrated open-source packages directly into physical robot-in-the-loop (HIL/SIL) testing environments.

Software Development Manager – Comau 03.2016 – 10.2017

Directed the technical software design, architecture, and deployment of Comau's next-generation industrial robotics platform across PC, mobile, and real-time controller environments.

Translated ambiguous real-world electromechanical behaviors into tractable, safe, and robust motion control models.

Led cross-functional engineering teams, fostering technical development, cross-department alignment, and execution of complex project milestones.

Software Engineer Specialist in Motion Planning and Robotics – Comau 03.2007 – 02.2016

Designed, coded, and validated advanced motion control algorithms, kinematic solvers, and trajectory planners for Comau industrial manipulators using C/C++ on Real-Time Operating Systems (RTOS).

Active technical contributor to high-profile EU Projects (ROBOFOOT, SMErobotics, X-Act); developed the C4G/C5G Open and Open Realistic Robot Library to support motion planning on open Linux platforms.

Led the internal eMotion project, directly optimizing multi-body dynamic models to maximize robot velocity profile smoothness while reducing mechanical stress.

Engineered engineering-grade software solutions for advanced Conveyor Tracking and contact/collision avoidance in overlapping Interference Regions, managing complex flexible assemblies and multi-robot workspace safety.

Software Engineer Specialist in Motion Planning and Robotics – ICAPGroup 10.2006 – 03.2007

Integrated highly automated, safety-critical robotic cells for the automotive manufacturing sector using Comau and ABB platforms.

Designed specialized computer vision systems using deterministic programming languages and programmed complex Allen-Bradley PLC and Pilz hardware-in-the-loop security configurations.

Activities, achievements & certifications

- Associate Editor Online Journal of Robotics & Automation Technology (OJRAT)
- Editorial Board Member Sciencefather
- Graduated with highest honors from Rome University of Tor Vergata
- Reviewer: IEEE Sensors Journal, ICRA 2023, IEEE Journal of Radio Frequency Identification, Elsevier Computers & electrical engineering, Elsevier Digital signal processing
- Second prize winner at EUnited Robotics Technology Transfer Award 2014

- Finalist at EUROP/EURON Technology Transfer Award 2009
- Second prize at Altran 'Campus' Award 2006
- Fundamentals of accelerated computing with CUDA Python – NVIDIA certification
- Disaster risk monitoring using satellite imagery – NVIDIA certification
- Getting started with AI on Jetson Nano – NVIDIA certification
- Building Video AI applications at the edge on Jetson Nano – NVIDIA certification
- Develop, Customize, and publish in Omniverse with extensions – NVIDIA certification

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Skills and competences

Core Research & Systems: Physics-Informed ML, World Models, Sim-to-Real Alignment, Multi-Body Dynamics, Synthetic Data Generation, Contact & Friction Modeling, Sensor Fusion.
Programming Languages: Python, C, C++, MATLAB, JavaScript, Perl, FORTRAN.
Frameworks, Libraries & Tools: PyTorch, TensorFlow, ROS/ROS2, OpenUSD, OpenCV, NumPy, Pandas, GitLab CI/CD, AWS Cloud Infrastructure.
Languages: Italian (Native), English (Full professional proficiency), Spanish (Full professional proficiency), French (Limited working proficiency).

Selected publications

Romanelli, F., & Martinelli, F. (2023). *Synthetic sensor measurement generation with noise learning and multi-modal information*. IEEE Access, doi: 10.1109/ACCESS.2023.3323038.

Romanelli, F., & Martinelli, F. (2023). *Synthetic Sensor Data Generation Exploiting Deep Learning Techniques and Multi-Modal Information*. IEEE Sensors Letters, vol. 7, no. 7, pp. 1-4, July 2023, Art no. 7003404, doi: 10.1109/LSENS.2023.3290209.

Romanelli, F., Martinelli, F., & Mattogno, S. (2023). *Resilient simultaneous localization and mapping fusing Ultra Wide Band range measurements and Visual Odometry*. J Intell Robot Syst 109, 64 (2023). https://doi.org/10.1007/s10846-023-01995-z

Navone, A., Romanelli, F., Ambrosio, M., Martini, M., Angarano, S., & Chiaberge, M. (2023). *Lavender Autonomous Navigation with Semantic Segmentation at the Edge*. In: Meo, R., Silvestri, F. (eds) Machine Learning and Principles and Practice of Knowledge Discovery in Databases. ECML PKDD 2023. Communications in Computer and Information Science, vol 2135. Springer, Cham. https://doi.org/10.1007/978-3-031-74633-8_18.

Bianchi, L., Carnevale, D., Del Frate, F., Masocco, R., Mattogno, S., Romanelli, F., & Tenaglia, A. (2023). *A novel distributed architecture for unmanned aircraft systems based on Robot Operating System 2*. IET Cyber-Systems and Robotics, 5(1), e12083.

Martinelli, F., Mattogno, S., & Romanelli, F. (2023). *A resilient solution to Range-Only SLAM based on a decoupled landmark range and bearing reconstruction*. Robotics and Autonomous Systems, 160, 104324.

Bianchi, L., Carnevale, D., Masocco, R., Mattogno, S., Oliva, F., Romanelli, F., & Tenaglia, A. (2023). *Efficient visual sensor fusion for autonomous agents*. In 2023 International Conference on Control, Automation and Diagnosis (ICCAD) (pp. 01-06). IEEE.

Romanelli, F., Martinelli, F., & Di Giampaolo, E. (2022). *Robust simultaneous localization and mapping using range and bearing estimation of radio ultra high frequency identification tags*. IEEE Transactions on Control Systems Technology, 31(2), 772-785.

Fenucci, A., Indri, M., & Romanelli, F. (2014). *A real time distributed approach to collision avoidance for industrial manipulators*. In Proceedings of the 2014 IEEE Emerging Technology and Factory Automation (ETFA) (pp. 1-8).

Michieletto, S., Tosello, E., Romanelli, F., Ferrara, V., & Menegatti, E. (2014). *ROS-I interface for COMAU robots*. In Simulation, Modeling, and Programming for Autonomous Robots: 4th International Conference, SIMPAR 2014, Bergamo, Italy, October 20-23, 2014. Proceedings 4 (pp. 243-254). Springer International Publishing.

Bascetta, L., Ferretti, G., Magnani, G., Rocco, P., Abba, F., Gerio, G. P., & Romanelli, F. *Towards an industrial implementation of the walk-through programming technique for robotic manipulators*. ICRA 2012 IEEE International Conference on Robotics and Automation.
Antonelli, G., Chiavellini, S., Perna, V., Romanelli, F., *A Modular and Task-oriented Architecture for Open Control System: the Evolution of C5G Open towards High Level Programming*, The ICRA 2010 Workshop on Innovative Robot Control Architectures for Demanding (Research) Applications, Lund University, Sweden, Nov 2010.

Patents

- Method for controlling at least two robots having respective working spaces including at least one region in common - US US20130110288 A1 · Issued May 2, 2013
- Robot system - EU EP 2194434 (A1) · Issued Dec 5, 2008 and US US 8,412,379 B2 · Issued Apr 2, 2013